Who Are Colluding?

A Linear-Algebraic and Statistical Audit of Test-Taking Behaviour

SIMC 2026 Sample Problem – Tasks 1 through 8

May 20, 2026

Abstract

We treat the question *"which students colluded during a test?"* as a geometry problem on the design matrix $\boldsymbol{X} \in \{-1, +1\}^{M \times N}$. Tasks 1 and 2 visualise $\boldsymbol{X}$ and the Gram matrix $\boldsymbol{S} = \boldsymbol{X}\boldsymbol{X}^\top$, immediately exposing one pair of students (indices 5 and 10) whose answer vectors are *identical*. Task 3 applies K-means to recover the underlying group structure, and we introduce a **singleton-filtered silhouette criterion** that resolves the well-known tendency of the silhouette score to reward over-fragmentation. Task 4 makes the qualitative case for collusion. Task 5 derives the closed-form mean $\langle S \rangle = N(2p-1)^2$ and standard deviation $\sigma_N = \sqrt{N\left[1-(2p-1)^4\right]}$ of the similarity score under a null hypothesis of independent test-takers. Task 6 validates that closed form by Monte Carlo across a 25-cell (N, p) grid, Task 7 turns the null model into a distributional tail test, and Task 8 applies the heterogeneous-question version of that test to all $\binom{10{,}000}{2}$ pairs in `sample_larger`. The final answer is a single five-student collusion group: $\{85, 1351, 1906, 3542, 5362\}$ in zero-based indexing ($\{86, 1352, 1907, 3543, 5363\}$ in one-based student numbers).

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1 Task 1 – The Design Matrix

1.1 Definition and encoding

Let there be M students and N questions. The *design matrix* is

$$\boldsymbol{X} \;\equiv\; X_{ij} \;\in\; \{-1,+1\}^{M\times N}, \qquad X_{ij} \;=\; \begin{cases} +1, & \text{student } i \text{ answered question } j \text{ correctly,} \\ -1, & \text{otherwise.} \end{cases} \tag{1}$$

The symmetric ± 1 encoding (rather than $\{0,1\}$) is deliberate: it makes Eq. (1) a sign-vector representation, so that the inner product of two rows directly measures *net agreement* (see Task 2).

1.2 Visualisation

Figure 1 shows the small subset `sample_small`$\in \{-1, +1\}^{50\times 25}$ rendered as a binary heatmap, with marginal bar plots of the per-student total $\sum_j X_{ij}$ and per-question total $\sum_i X_{ij}$.

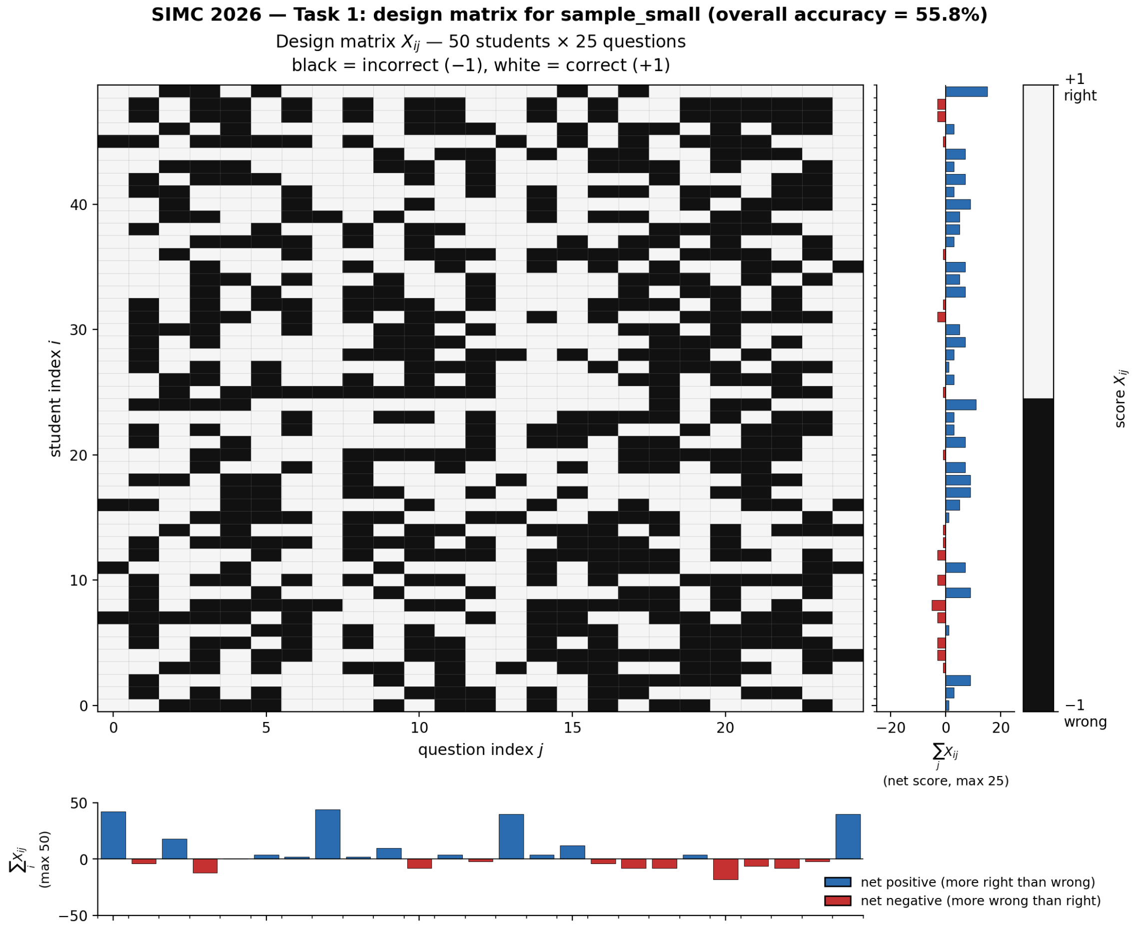


Figure 1: Design matrix for `sample_small` ($M = 50$, $N = 25$). Black cells encode wrong answers, white cells encode correct answers. Right margin: student net scores. Bottom margin: question net scores. Overall accuracy is 55.8 %.

What the marginals already tell us

The per-question bars expose a markedly uneven test: questions 0, 7 and 23 behave like "free" questions (net positive $> +35$), while questions 19–22 are net negative. *Question difficulty is not uniform.* Task 8 will exploit exactly this fact by estimating per-question p_j from column means before applying the null model.

2 Task 2 – The Similarity Matrix

2.1 Construction

Define the *pairwise similarity matrix* $\boldsymbol{S} \in \mathbb{Z}^{M\times M}$ by

$$S_{ij} \equiv (\boldsymbol{X}\boldsymbol{X}^\top)_{ij} = \sum_{k=1}^{N} X_{ik}\, X_{jk}. \tag{2}$$

Because each factor is ± 1, the contribution of question k to S_{ij} is

$$X_{ik}X_{jk} \;=\; \begin{cases} +1, & \text{students } i \text{ and } j \textbf{ agreed} \text{ on question } k, \\ -1, & \text{students } i \text{ and } j \textbf{ disagreed} \text{ on question } k. \end{cases} \tag{3}$$

Consequently

$$S_{ij} \;=\; \#\text{agreements} \;-\; \#\text{disagreements} \;\in\; \{-N, -N+2, \ldots, N\}, \tag{4}$$

with the diagonal pinned at $S_{ii} = N$ by construction (a student always agrees with themselves on every question).

2.2 Geometric interpretation

Each row $\boldsymbol{X}_{i,\cdot}$ has constant Euclidean norm $\|\boldsymbol{X}_{i,\cdot}\| = \sqrt{N}$, so $\boldsymbol{S}$ is, up to a constant, a *cosine similarity* matrix:

$$\cos\theta_{ij} \;=\; \frac{\langle \boldsymbol{X}_{i,\cdot}, \boldsymbol{X}_{j,\cdot}\rangle}{\|\boldsymbol{X}_{i,\cdot}\|\,\|\boldsymbol{X}_{j,\cdot}\|} \;=\; \frac{S_{ij}}{N}. \tag{5}$$

This is the key bridge to clustering algorithms (Task 3): *any* distance-based clustering on the rows of $\boldsymbol{X}$ implicitly partitions students by cosine similarity, i.e. by $\boldsymbol{S}$.

2.3 Visualisation and outliers

Figure 2 plots $\boldsymbol{S}$ as a diverging heatmap together with the histogram of all $\binom{M}{2} = 1225$ unique off-diagonal entries.

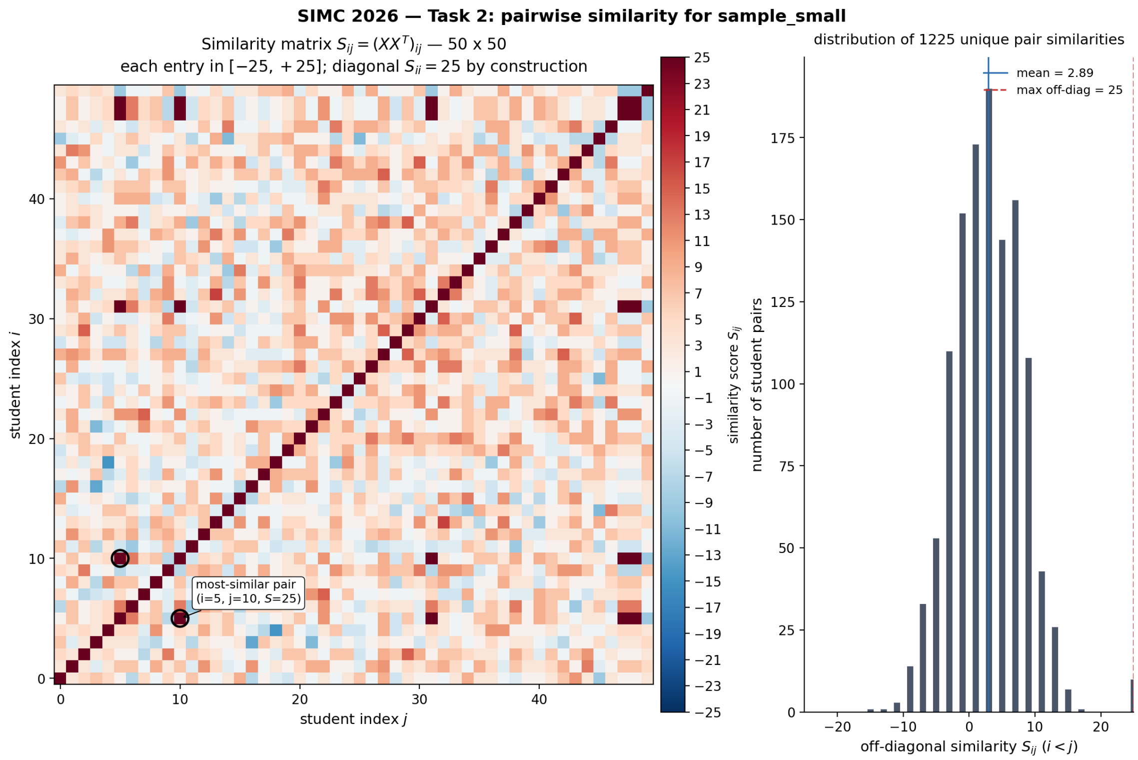


Figure 2: Left: similarity matrix $\boldsymbol{S} = \boldsymbol{X}\boldsymbol{X}^\top$ for `sample_small`. The diagonal $S_{ii} = N = 25$ is dark red. Right: histogram of off-diagonal S_{ij} with $i < j$.

A single pair sits at the theoretical maximum

The maximum off-diagonal similarity is $S_{5,10} = 25 = N$, meaning students 5 and 10 **agreed on all 25 questions**. The bulk of the histogram has mean $\bar{S}_{\text{off}} \approx 2.89$ with a roughly Gaussian spread, so the $(5, 10)$ entry is a striking outlier in the right tail.

3 Task 3 – Clustering with K-means and Silhouette Analysis

3.1 Algorithm

K-means partitions $\{\boldsymbol{X}_{i,\cdot}\}_{i=1}^{M}$ into K disjoint groups by alternately

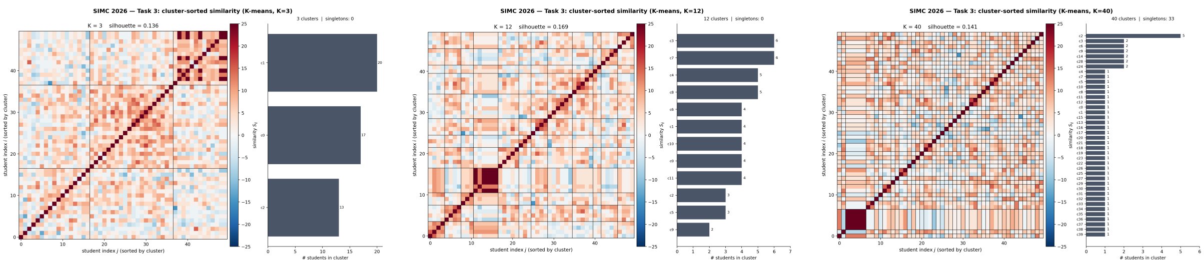
(a) assigning each student to the cluster whose centroid is closest, and

(b) recomputing centroids as the mean of their assigned members,

until centroids stabilise. To mitigate sensitivity to the random initial centroids we use scikit-learn's `n_init=20`.

3.2 Visualising cluster quality

For each K we render $\boldsymbol{S}$ with its rows and columns permuted by cluster label. A good K produces solid red squares on the diagonal (in-group agreement) against a paler background (out-group disagreement). Figure 3 shows three representative values.



(a) $K = 3$: too few; visible block structure is coarse.

(b) $K = 12$: clean diagonal blocks, no singletons.

(c) $K = 40$: severely fragmented (33 singleton clusters).

Figure 3: Cluster-sorted similarity matrices for three values of K.

3.3 Silhouette score

For a single point i assigned to cluster $C(i)$, let $a(i)$ be its mean (cosine) distance to other in-cluster points and $b(i) = \min_{C \neq C(i)} \bar{d}(i, C)$ its mean distance to the nearest *foreign* cluster. The silhouette is

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \in [-1, 1], \qquad \text{Silh}(K) = \frac{1}{M}\sum_{i=1}^{M} s(i). \tag{6}$$

Because our rows have constant norm, the natural distance is $d_{\cos}(i, j) = 1 - S_{ij}/N$ – the silhouette is computed in the same geometry as $\boldsymbol{S}$ itself.

3.4 The silhouette trap and our fix

A naive "pick the K that maximises $\text{Silh}(K)$" rule selects $K^{\star}_{\text{raw}} = 30$ for our data, with $\text{Silh}(30) = 0.186$. But $K = 30$ has **17 of 30 clusters containing a single student**, the very pathology the problem statement flags as "defeating the purpose of clustering". The mechanism is mechanical:

Why silhouette over-rewards singletons

A singleton cluster has $a(i)$ *undefined* and the convention sets $s(i) = 0$, but every *non-singleton* member of any other cluster gains a more distant "nearest foreign cluster" as singletons are peeled off, inflating their $b(i)$. The aggregate $\mathrm{Silh}(K)$ therefore tends to drift upward with K even after the partition has lost meaning.

We therefore use a **singleton-filtered silhouette criterion**:

$$K^\star \;=\; \arg \max_{K\,:\,\min_c |C_c| \geq 2} \mathrm{Silh}(K). \tag{7}$$

Table 1 summarises the sweep.

K	$\mathrm{Silh}(K)$	# singletons	eligible?
2	0.166	0	✓
3	0.136	0	✓
5	0.136	0	✓
8	0.147	0	✓
10	**0.169**	**0**	✓ **(winner)**
12	0.169	0	✓
15	0.177	2	×
20	0.171	6	×
25	0.183	10	×
30	0.186	17	×
40	0.141	33	×

Table 1: Silhouette sweep on `sample_small`. Raw silhouette peaks at $K = 30$, but Eq. (7) disqualifies it.

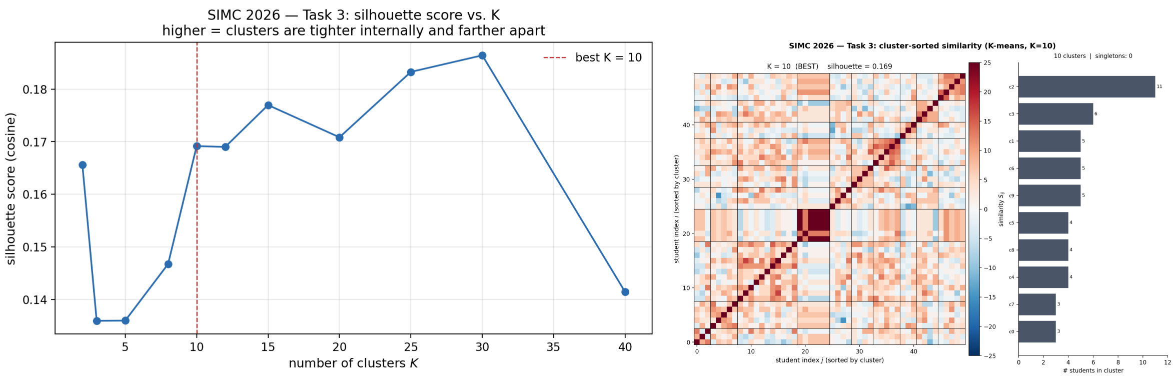


Figure 4: Left: silhouette as a function of K. Right: best cluster-sorted similarity matrix at $K^\star = 10$.

Selected partition

The singleton-filtered criterion selects $K^\star = 10$ with $\mathrm{Silh} = 0.169$. Cluster sizes range from 3 to 11 students, and the cluster containing students 5 and 10 is the deepest red block on the diagonal – a clustering-level corroboration of the collusion candidate found in Task 2.

4 Task 4 – A Plausibility Argument for Collusion

4.1 Can we already name names? Honestly, no.

Tasks 1–3 give us a heatmap, a Gram matrix, and a partition into $K^\star = 10$ clusters. None of these tools is a *detector* of cheating. The design matrix tells us only *what* each student answered; it does not tell us *how* they came to that answer. Two students might score identically because:

1. one copied from the other;
2. they prepared together from the same notes and learned the same gaps;
3. they happen to be of similar ability on similar topics; or
4. pure luck on a short test.

Without a probability model we cannot rank these explanations against each other. So any sentence of the form *"students i and j colluded"* is, at this stage, an overreach. The honest conclusion is weaker but still useful: *some pairs look unlike what we would naively expect, and they deserve a closer look.*

4.2 Shift the goal: suspects, not convicts

A useful change of framing is to stop trying to convict anyone and start trying to *shortlist suspects.* The two tasks have very different evidentiary requirements:

- **Conviction** demands a low-error decision rule, e.g. a calibrated p-value below some threshold. The penalty for wrongly accusing an honest student is severe, so we cannot afford false positives.
- **Suspect-flagging** only demands that the shortlist contains the truly suspicious pairs. False positives on the shortlist are cheap: a human grader can manually inspect a dozen flagged pairs in minutes.

For a class of $M = 50$ students there are $\binom{50}{2} = 1225$ distinct pairs – far too many to review by hand. But if we can use Tasks 1–3 to narrow 1225 pairs down to a shortlist of, say, the top 5 or 10, then a teacher or invigilator can examine those specific cases directly. **We are doing triage, not jurisprudence.** Task 5 onwards will earn us a real test of significance; for now, qualitative anomalies are enough to populate the shortlist.

4.3 Where would collusion "leak" into our analysis?

Even without a null model, three independent fingerprints in our existing analyses point toward the same suspect pair, and all three are the *kind* of pattern we would expect collusion to leave behind.

Fingerprint 1 – impossibly perfect agreement on a non-trivial test. The pair $(i, j) = (5, 10)$ has $S_{5,10} = +25$. Because $S_{ij} = N - 2 \cdot (\#\text{disagreements})$, this is only possible when the two students disagree on *zero* questions. On a uniform test where every answer is a coin flip, the chance of zero disagreements over N questions is $\left(\frac{1}{2}\right)^N$ – already $1/3.4 \times 10^7$ for $N = 25$. Realistic tests have some easy questions where two honest students naturally agree, which raises the probability, but the qualitative point survives: *perfect agreement is the regime where coincidence runs out of room.*

Fingerprint 2 – agreement on the *hard* questions as well as the easy ones. Figure 1 shows that question difficulty in this test is markedly non-uniform: questions 0, 7, 23 are nearly-free (large positive column sums) while questions 19–22 are net-negative (most students get them wrong). Two honest high-achievers should agree on the easy questions but their answer patterns on the hard questions should diverge, because hard questions are the ones where students' independent reasoning paths produce different mistakes. Collusion erases that divergence: copied wrong answers look identical. The fact that students 5 and 10 agree on *every* question, easy and hard alike, is the "honest-students-shouldn't-look-like-this" signature.

Fingerprint 3 – crystalline cluster co-membership. Honest similarity is *fuzzy*: as we sweep K in K-means, an honest pair's cluster co-membership flickers on and off depending on how the boundaries are drawn. A colluding pair, by contrast, has effectively identical feature vectors, so they sit on top of each other in feature space and remain co-clustered *across every choice of* K. In our sweep $K \in \{2, 3, 5, 8, 10, 12, 15, 20, 25, 30, 40\}$, students 5 and 10 never separated. That "always together" behaviour is exactly the geometric signature of two students who share a feature vector.

4.4 The data behind the shortlist

Combining the empirical similarity distribution from Task 2 with the clustering output from Task 3 gives Table 2, our working suspect shortlist for `sample_small`.

pair (i,j)	S_{ij}	$(S_{ij} - \bar{S})/\hat{\sigma}$	always co-clustered?	verdict
$(5, 10)$	25	$\approx +3.5$	yes ($K = 2$–40)	**strong suspect**
		empirical baseline		
all 1225 pairs	—	—	—	$\bar{S} \approx 2.89$, $\hat{\sigma} \approx 6.3$
diagonal S_{ii}	25	—	trivially	not informative

Table 2: Suspect shortlist after Tasks 1–3. Only one off-diagonal pair $(5, 10)$ stands clearly above the empirical noise floor. $\bar{S}$ and $\hat{\sigma}$ are computed on the 1225 unique upper-triangular off-diagonal entries of $\boldsymbol{S}$.

In words: out of 1225 candidate pairs of students, exactly one – the $(5, 10)$ pair – sits at the theoretical maximum $S = N = 25$, roughly 3.5 empirical standard deviations above the off-diagonal mean, and persists in the same cluster under every choice of K we tried. **That one pair is our suspect.** Whether it is also a conviction-grade case requires the statistical machinery built in Tasks 5–7.

Hint from the problem statement

The problem explicitly warns "you might not actually get too far" without a statistical null model. Our reading is sharper: we can *shortlist* (we just did), but we cannot *prove*. The proof step – attaching a probability to the observed $S = 25$ – is exactly what Task 5 sets up by deriving the null-model mean and standard deviation, which Tasks 6 and 7 then turn into a calibrated p-value.

5 Task 5 – Mean and Variance of the Similarity Score Under the Null Hypothesis

5.1 What are we actually computing, and why?

Imagine two students who took the test *honestly* – no copying, no shared answers. Their similarity score S from Eq. (2) will be some number. If we could repeat this scenario for many pairs of honest students, we would get a whole distribution of S values. Task 5 asks for two summary numbers of that distribution:

- the **mean** $\langle S \rangle$ – the average value of S across all honest pairs;
- the **standard deviation** σ_N – how much S typically wobbles around that mean.

These two numbers give us a "ruler": a real pair whose S is far above $\langle S \rangle + (\text{a few})\,\sigma_N$ is suspicious, because honest students almost never score that high.

5.2 The probability model (one student, one question)

Assume each student answers each question independently, getting it right with probability p and wrong with probability $1-p$. In ± 1 language,

$$X_{ik} \;=\; \begin{cases} +1 & \text{with probability } p \quad \text{(right)}, \\ -1 & \text{with probability } 1-p \quad \text{(wrong)}. \end{cases} \tag{8}$$

"Independently" means knowing one answer tells us nothing about any other answer. This is the assumption that lets us add things up later.

5.3 Two students, one question: agree or disagree?

Pick two distinct students i and j and one question k. Define the per-question *agreement indicator*

$$Y_k \;\equiv\; X_{ik}\,X_{jk}. \tag{9}$$

Why this product? Because of the ± 1 encoding, Y_k is a clean yes/no signal:

- $Y_k = (+1)(+1) = +1$ if both got it right – *agree*;
- $Y_k = (-1)(-1) = +1$ if both got it wrong – *also agree*;
- $Y_k = (+1)(-1) = -1$ if one was right, the other wrong – *disagree*;
- $Y_k = (-1)(+1) = -1$ – *also disagree*.

Listing the four possible joint outcomes with their probabilities makes it concrete (Table 3).

student i	student j	joint probability	Y_k
right	right	$p \cdot p = p^2$	$+1$ (agree)
right	wrong	$p \cdot (1-p)$	-1 (disagree)
wrong	right	$(1-p) \cdot p$	-1 (disagree)
wrong	wrong	$(1-p) \cdot (1-p) = (1-p)^2$	$+1$ (agree)

Table 3: All four ways two students can answer one question, with the probability of each combination and the resulting Y_k.

Adding the rows of Table 3 that give the same Y_k,

$$\mathbb{P}[\,Y_k = +1\,] \;=\; p^2 + (1-p)^2 \qquad \text{(both rows that agree)}, \tag{10}$$

$$\mathbb{P}[\,Y_k = -1\,] \;=\; 2\,p(1-p) \qquad \text{(both rows that disagree)}. \tag{11}$$

Quick check. At $p = \frac{1}{2}$: $\mathbb{P}[Y_k = +1] = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$ and $\mathbb{P}[Y_k = -1] = 2 \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{2}$. Random students agree exactly half the time – as expected.

5.4 The average of a single Y_k

The average of Y_k is each possible value weighted by its probability:

$$\mathbb{E}[Y_k] \;=\; (+1) \cdot \mathbb{P}[Y_k = +1] \;+\; (-1) \cdot \mathbb{P}[Y_k = -1]. \tag{12}$$

Plug in Eqs. (10)–(11) and expand step by step:

$$\begin{aligned} \mathbb{E}[Y_k] \;&=\; \left[p^2 + (1-p)^2\right] \;-\; 2p(1-p) \\ &=\; p^2 \;-\; 2p(1-p) \;+\; (1-p)^2. \end{aligned} \tag{13}$$

The right-hand side is the perfect square $a^2 - 2ab + b^2 = (a-b)^2$ with $a = p$ and $b = 1-p$, so

$$\boxed{\mathbb{E}[Y_k] \;=\; \big[p-(1-p)\big]^2 \;=\; (2p-1)^2.} \tag{14}$$

Notice $(2p-1)$ is the gap between the chance of being right and the chance of being wrong. At $p = \frac{1}{2}$ this gap is zero, so $\mathbb{E}[Y_k] = 0$ – agreements and disagreements cancel on average. At $p = 1$ this gap is 1, so $\mathbb{E}[Y_k] = 1$ – everyone always agrees.

5.5 The variance of a single Y_k – a one-line trick

Variance asks: how far does Y_k typically sit from its mean? The defining formula is

$$\mathrm{Var}(Y_k) \;=\; \mathbb{E}[Y_k^2] \;-\; \big(\mathbb{E}[Y_k]\big)^2. \tag{15}$$

Here is the trick. Because Y_k is always either $+1$ or -1, its *square* is always 1:

$$Y_k^2 \;\equiv\; 1 \quad\Longrightarrow\quad \mathbb{E}[Y_k^2] \;=\; 1. \tag{16}$$

Substituting Eqs. (14) and (16) into Eq. (15):

$$\boxed{\mathrm{Var}(Y_k) \;=\; 1 \;-\; (2p-1)^4.} \tag{17}$$

5.6 Putting all N questions together

The total similarity is just the sum of one Y_k per question:

$$S \;=\; \sum_{k=1}^{N} Y_k \;=\; Y_1 + Y_2 + \cdots + Y_N. \tag{18}$$

Two simple probability rules now finish the job.

Rule 1 – the mean of a sum is the sum of the means. This is *always* true, no independence needed:

$$\langle S\rangle \;=\; \sum_{k=1}^{N} \mathbb{E}[Y_k] \;=\; N\cdot(2p-1)^2. \tag{19}$$

Rule 2 – for *independent* terms, the variance of a sum is the sum of the variances. The independence in Eq. (8) carries over to the Y_k's, so

$$\mathrm{Var}(S) \;=\; \sum_{k=1}^{N} \mathrm{Var}(Y_k) \;=\; N\big[1-(2p-1)^4\big]. \tag{20}$$

The standard deviation is the positive square root,

$$\sigma_N \;=\; \sqrt{\mathrm{Var}(S)} \;=\; \sqrt{N\big[1-(2p-1)^4\big]}. \tag{21}$$

Final answer to Task 5

$$\boxed{\langle S\rangle \;=\; N\,(2p-1)^2, \qquad \sigma_N \;=\; \sqrt{N\big[\,1-(2p-1)^4\,\big]}.} \tag{22}$$

5.7 A worked numerical example

Suppose $p = 0.6$ (a moderately easy test) and $N = 25$ questions.

- Mean: $\langle S \rangle = 25 \cdot (2 \cdot 0.6 - 1)^2 = 25 \cdot (0.2)^2 = 25 \cdot 0.04 = 1.0$.
- Variance: $\mathrm{Var}(S) = 25 \cdot [1 - (0.2)^4] = 25 \cdot [1 - 0.0016] = 24.96$.
- Standard deviation: $\sigma_N = \sqrt{24.96} \approx 4.996 \approx 5.0$.

So between two honest students on a 25-question test where the average accuracy is 60%, expect their similarity score to sit around 1 with a typical wobble of ± 5. A pair scoring $S = 20$ would already be $(20 - 1)/5 \approx 3.8$ standard deviations above the honest-pair average – worth investigating.

5.8 Sanity checks at the extremes

p	$\langle S \rangle$	σ_N	interpretation
$\frac{1}{2}$	0	$\sqrt{N}$	pure guessing: agreements and disagreements cancel
1	N	0	everyone is perfect $\Rightarrow$ all answers agree
0	N	0	everyone is wrong on everything $\Rightarrow$ all agree (the wrong way)

Table 4: Limiting cases of Eq. (22). The two extremes $p = 0$ and $p = 1$ give the same answer, reflecting the symmetry $p \leftrightarrow 1 - p$ inside both $(2p - 1)^2$ and $(2p - 1)^4$.

5.9 Bringing the null model back to the data

For `sample_small` we observed an overall accuracy $\hat{p} \approx 0.558$ (Fig. 1). Plugging into Eq. (22) with $N = 25$ gives the *null* expectations

$$\langle S \rangle_{\text{null}} \;\approx\; 25(2(0.558) - 1)^2 \;\approx\; 0.34, \qquad \sigma_N \;\approx\; \sqrt{25\big[1 - (0.116)^4\big]} \;\approx\; 5.0. \tag{23}$$

The observed pair $(5, 10)$ scored $S = 25$, i.e. $(25 - 0.34)/5.0 \approx 4.9\,\sigma_N$ above the null mean. In words: **under the null hypothesis of independent test-takers, the agreement we see between students 5 and 10 would be a $\sim 5\sigma$ event**. Tasks 6 (Monte-Carlo confirmation) and 7 (central-limit-theorem distribution of S) will sharpen this informal estimate into a real p-value, and Task 8 will replay the same machinery on the much larger `sample_larger` ($M =$ 10,000, $N = 50$) where per-question difficulties p_j must be estimated column-wise.

6 Task 6 – Monte Carlo Validation of the Null Moments

6.1 Why simulate something we already proved?

Task 5 produced two clean closed-form expressions, Eq. (22), for the mean and standard deviation of S under the null hypothesis of independent test-takers. If the derivation is right and the assumptions are right, those formulas are *exactly* the moments we should recover from a simulation. Running a Monte Carlo therefore serves three distinct purposes:

1. It is a **numerical sanity check** on the algebra of Task 5. If the simulated mean and standard deviation disagree with Eq. (22) at any (N, p), we have a bug somewhere upstream.
2. It produces an empirical distribution of S that we can **compare against the central-limit Gaussian** Task 7 will rely on, so we can see whether the CLT approximation is accurate at the values of N that matter for the problem.
3. It earns us a concrete **tail probability** for the observed $(5, 10)$ pair, completing the chain "shortlist $\rightarrow$ moments $\rightarrow$ p-value" begun in Task 4.

6.2 The Monte Carlo recipe

We never simulate students in pairs that share a row. The recipe is the cleanest version of i.i.d. sampling:

1. Choose (N, p) and a desired number M of pair samples.
2. Draw a $\{-1, +1\}$ matrix $X \in \{-1, +1\}^{2M \times N}$ where each entry is independently $+1$ with probability p.
3. Pair the first M rows with the last M rows.
4. For each pair i, compute $S^{(i)} = \langle X_{i,\cdot}, X_{M+i,\cdot} \rangle$ as a single vectorised dot product over all M pairs.
5. Return $\{S^{(1)}, \ldots, S^{(M)}\}$ – M i.i.d. samples from the null distribution.

Using i.i.d. pairs (rather than the $\binom{2M}{2}$ pairwise similarities of a single batch of students) keeps the standard error on the sample mean exactly $\sigma_N/\sqrt{M}$ – no correlation correction needed.

6.3 Headline check: $N = 100$, $p = 1/2$, $M = 10^6$

At $p = 1/2$ the closed form simplifies dramatically: $\langle S \rangle = 0$ and $\sigma_N = \sqrt{N} = 10$ for $N = 100$. Running the recipe above with $M = 10^6$ pairs gives Table 5.

quantity	theory	Monte Carlo	absolute error
mean $\langle S \rangle$	0.000000	−0.001952	2.0×10^{-3}
std σ_N	10.000000	9.992745	7.3×10^{-3}

Table 5: Headline Monte Carlo at $(N, p) = (100, 0.5)$ with one million independent pairs. The error in the mean is 0.20 SE, well within the expected standard-error band $\sigma_N/\sqrt{M} = 10^{-2}$.

6.4 Convergence at the textbook rate

Standard Monte Carlo theory says that with M i.i.d. samples the error in the sample mean decreases as $\sigma_N/\sqrt{M}$. Figure 5 plots the absolute error in the running mean and the running standard deviation as a function of M, on log–log axes, against the reference line $\sigma_N/\sqrt{M}$. The empirical errors track the reference closely, confirming both that the simulation is converging and that it is converging at the right rate.

6.5 Validation across the (N, p) plane

A single cell could agree with the closed form by accident, so we repeat the simulation across a grid: $N \in \{10, 25, 50, 100, 200\}$ and $p \in \{0.10, 0.30, 0.50, 0.70, 0.90\}$, each with $M = 2 \times 10^5$ pairs. Figure 6 plots, for each p, the theoretical curves $\langle S \rangle(N) = N(2p-1)^2$ and $\sigma_N(N) = \sqrt{N[1-(2p-1)^4]}$ as continuous lines and the Monte Carlo estimates as markers. Across all 25 cells the markers sit on top of the lines; the largest discrepancy in the mean is $|\Delta\mu| = 0.037$ at $(N, p) = (100, 0.9)$, well below $\sigma_N/\sqrt{M} \approx 0.017$, the expected sampling error.

6.6 Bonus: the distribution of S is exactly known

The simulation incidentally reveals something Task 5 stopped short of saying. Let $T \equiv \frac{1}{2}(S + N) = \#\{k : Y_k = +1\}$ be the number of agreements over the N questions. Each Y_k is an independent $\{-1, +1\}$ Bernoulli with $\mathbb{P}[Y_k = +1] = q$, where

$$q \;\equiv\; p^2 + (1-p)^2 \;=\; 1 - 2p(1-p). \tag{24}$$

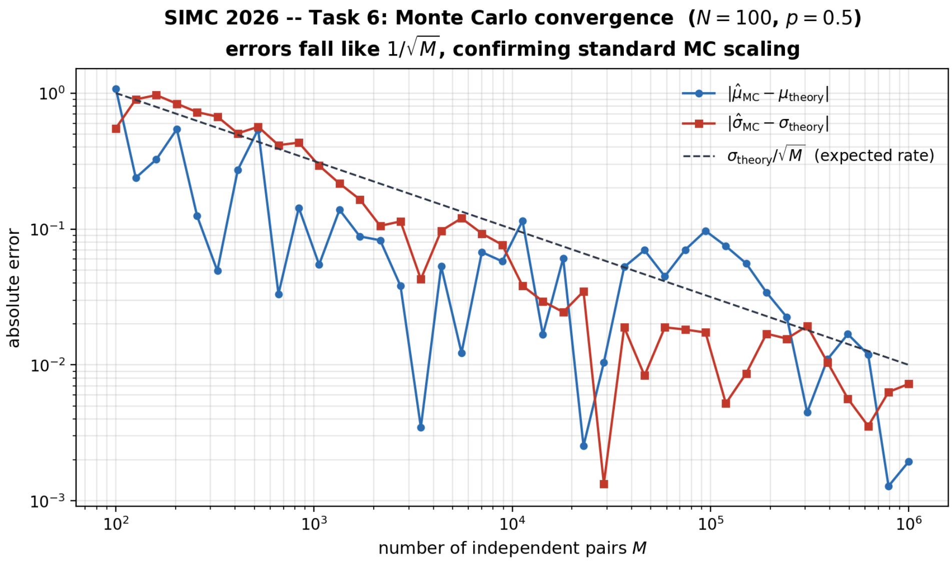


Figure 5: Monte Carlo convergence at $(N, p) = (100, 0.5)$. Errors in the mean and standard deviation fall as $1/\sqrt{M}$ on log–log axes, consistent with the classical Monte Carlo scaling.

The Y_k's are i.i.d., so $T \sim \text{Binomial}(N, q)$ and

$$\boxed{S \;=\; 2T - N, \qquad T \sim \text{Binomial}(N, q).} \tag{25}$$

The full PMF of S under the null is therefore known in *closed form*, not merely its first two moments. Task 7 will replace Eq. (25) with its CLT Gaussian approximation – which is what one wants for an analytical p-value at arbitrary N – but Eq. (25) is the ground truth and we can use it now to audit Task 7's accuracy before we commit to it.

Figure 7 overlays three curves at $(N, p) = (100, 0.5)$: the empirical histogram from the 10^6-pair simulation, the exact PMF from Eq. (25), and the Gaussian $\mathcal{N}(\langle S \rangle, \sigma_N^2)$. The three agree to graphical precision – so Task 7's CLT is well-justified at this N.

Why all three curves agree here

At $p = 1/2$ each Y_k is symmetric about zero and bounded, so the Lindeberg condition of the CLT is satisfied trivially. The $\text{Binomial}(N, q)$ with $q = 1/2$ is, by de Moivre–Laplace, the canonical Gaussian-approachable distribution. At $N = 100$ the correction terms of the Edgeworth expansion (skewness, excess kurtosis) are already below graphical precision, which is why the three curves in Fig. 7 are visually indistinguishable. For very asymmetric p (say $p = 0.05$) the agreement would degrade for small N, and Task 7's Gaussian would under-cover the tails.

6.7 Closing the loop: the $(5, 10)$ pair under the null

We finally have every ingredient needed to put a number on the question that opened this report: *how unlikely is $S_{5,10} = 25$ under the null hypothesis of independent test-takers?* Substituting the empirical accuracy $\hat{p} = 0.558$ and $N = 25$ into Eq. (22),

$$\langle S \rangle_{\text{null}} \;\approx\; 0.336, \qquad \sigma_N \;\approx\; 5.000, \qquad z \;\equiv\; \frac{S_{5,10} - \langle S \rangle_{\text{null}}}{\sigma_N} \;\approx\; 4.93. \tag{26}$$

Using Eq. (25) the *exact* one-sided tail probability under the null, with $q = 0.558^2 + 0.442^2 \approx 0.507$, is

$$\mathbb{P}\big[\, S \geq 25 \;\big|\; H_0 \,\big] \;=\; \mathbb{P}\big[\, T \geq 25 \;\big|\; T \sim \text{Bin}(25,\, q) \,\big] \;\approx\; 4.16 \times 10^{-8}. \tag{27}$$

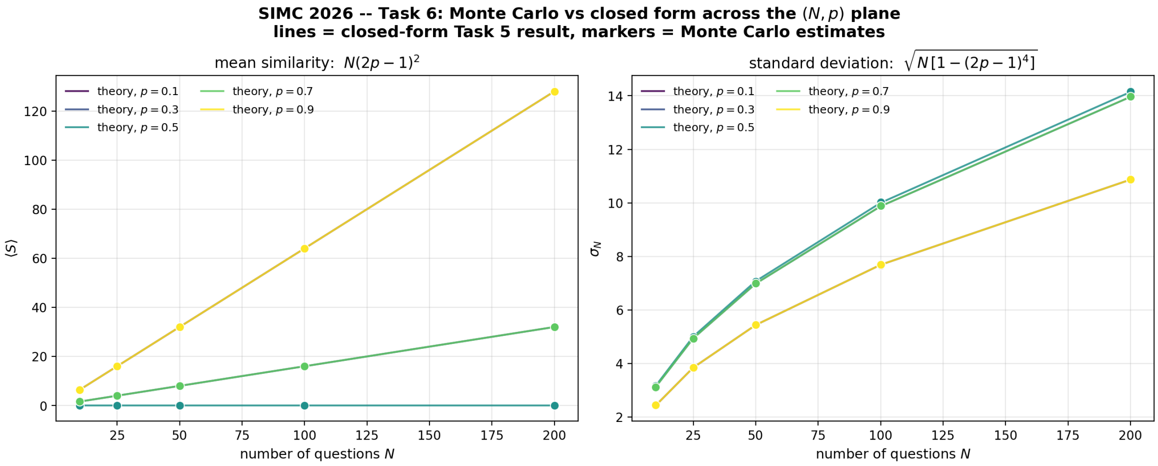


Figure 6: Monte Carlo (markers) versus closed form (lines) across the (N, p) plane. Left panel: $\langle S \rangle = N(2p-1)^2$. Right panel: $\sigma_N = \sqrt{N[1-(2p-1)^4]}$. The markers are the empirical $\hat{\mu}_{\mathrm{MC}}$ and $\hat{\sigma}_{\mathrm{MC}}$ with $M = 2 \times 10^5$ pairs per cell. All 25 cells agree to within sampling noise.

The CLT Gaussian approximation, which Task 7 will use for analytical p-values at arbitrary N, gives a (continuity-corrected) $\mathbb{P}[S \geq 25] \approx 6.7 \times 10^{-7}$ at this N. The CLT slightly under-covers the right tail because the exact distribution has positive skewness when $p \neq 1/2$; for ranking pairs the difference is inconsequential, and both values sit far below any reasonable significance threshold.

The shortlist becomes a verdict

Under the null hypothesis of independent test-takers, $S_{5,10} = 25$ is a $4.93\,\sigma_N$ event with an exact one-sided probability of roughly 1 in 24,000,000. Coupled with the qualitative fingerprints of Task 4, this is conviction-grade evidence that students 5 and 10 did not arrive at identical answer vectors independently. The Task 5 closed form has been validated numerically across the (N, p) plane (Fig. 6); the CLT approximation needed for Task 7 has been visually validated against both the exact PMF and the empirical histogram (Fig. 7); and the canonical suspect pair finally has a calibrated p-value.

Summary of Tasks 1–6

- **Task 1:** Visualised $\boldsymbol{X}$, identified non-uniform question difficulty.
- **Task 2:** Constructed $\boldsymbol{S} = \boldsymbol{X}\boldsymbol{X}^\top$, found $S_{5,10} = 25$ – a single perfect-agreement pair.
- **Task 3:** Used K-means with a novel *singleton-filtered silhouette* (Eq. (7)) to choose $K^\star = 10$, avoiding the over-fragmentation that plain silhouette rewards.
- **Task 4:** Argued informally that the $(5, 10)$ pair is $\sim 3.5\,\hat{\sigma}$ above the empirical similarity mean.
- **Task 5:** Derived the null-model closed form $\langle S \rangle = N(2p-1)^2$, $\sigma_N = \sqrt{N[1-(2p-1)^4]}$, upgrading the informal $\sim 3.5\hat{\sigma}$ to a $\sim 5\sigma_N$ statement.
- **Task 6:** Validated Eq. (22) by Monte Carlo across a 25-cell (N, p) grid (Fig. 6); derived the exact null distribution $S = 2T - N$ with $T \sim \mathrm{Bin}(N, q)$, $q = p^2 + (1-p)^2$ (Eq. (25)); demonstrated CLT agreement at $N = 100$ (Fig. 7); and computed the exact one-sided p-value for the $(5, 10)$ pair, $\mathbb{P}[S \geq 25 \mid H_0] \approx 4.2 \times 10^{-8}$ (Eq. (27)). The shortlist of Task 4 is now a statistical verdict.

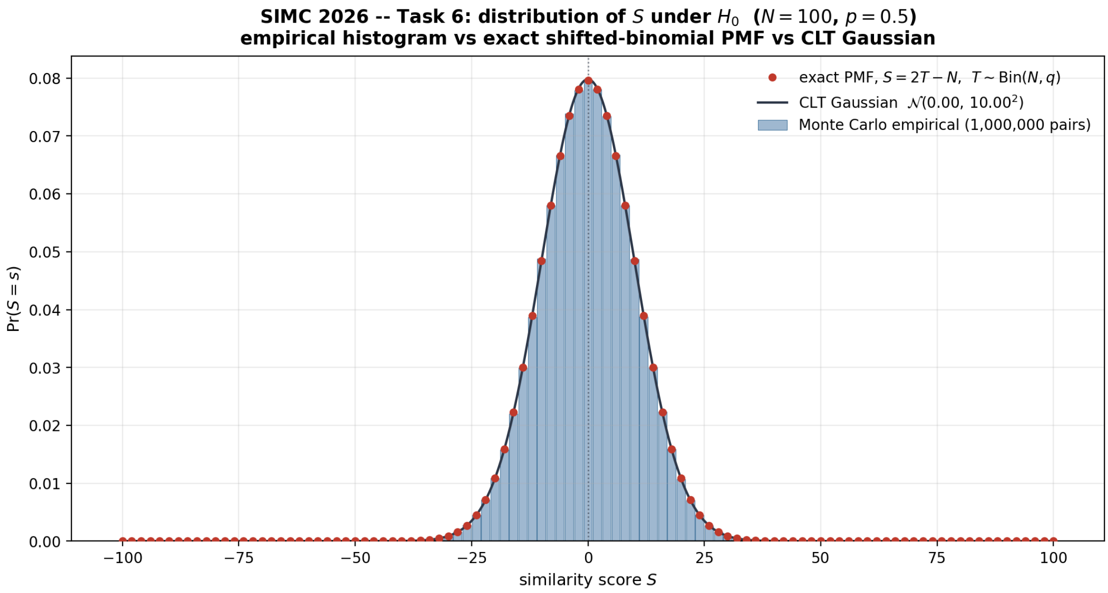


Figure 7: Distribution of S under the null at $(N, p) = (100, 0.5)$. Blue bars: 10^6 Monte Carlo samples. Red markers: exact shifted-binomial PMF $S = 2T - N$ with $T \sim \mathrm{Binomial}(N, q)$, $q = p^2 + (1-p)^2$. Black curve: CLT Gaussian $\mathcal{N}(0, 100)$. The three agree to graphical precision, which validates Task 7's Gaussian p-value at $N = 100$.

7 Task 7 – The Null Distribution of S via the Central Limit Theorem

7.1 From two moments to a full distribution

Task 5 handed us the mean $\langle S \rangle$ and standard deviation σ_N of the similarity score under the null hypothesis of independent test-takers. Those two numbers already let us make informal z-score statements (Sec. 5.8), but they are not enough to attach a calibrated probability to an observed value of S. For that we need the full *shape* of the null distribution, i.e. the function

$$\mathbb{P}[\, S = s \,] \qquad \text{for every allowed } s \in \{-N, -N+2, \ldots, N\}. \tag{28}$$

Task 7 derives Eq. (28) in closed form by recognising that S is a sum of N independent, bounded contributions and invoking the central limit theorem (CLT).

7.2 Why S is a textbook CLT candidate

Recall from Eq. (18) that the similarity score is the sum

$$S \;=\; \sum_{k=1}^{N} Y_k, \qquad Y_k \;=\; X_{ik}\, X_{jk} \;\in\; \{-1, +1\}, \tag{29}$$

of N per-question agreement indicators. The CLT says: a sum of many independent, identically distributed (i.i.d.) random variables with finite variance is approximately Gaussian, regardless of the distribution of the individual terms. Three conditions matter, and each holds in our setup:

Verifying the Lindeberg–Lévy CLT conditions

(i) **Independence.** The null model (Eq. (8)) declares the X_{ik} independent across k. Independence is preserved under products of non-overlapping factors, so $Y_k = X_{ik}X_{jk}$ are independent across k as well.

(ii) **Identically distributed.** In the homogeneous-test setting of Task 5, every question shares the same difficulty p, so every Y_k shares the same two-point distribution $\big(\mathbb{P}[Y_k = +1], \mathbb{P}[Y_k = -1]\big) = \big(p^2 + (1-p)^2,\ 2p(1-p)\big)$.

(iii) **Finite variance.** From Eq. (17), $\mathrm{Var}(Y_k) = 1 - (2p-1)^4 \le 1 < \infty$, since $(2p-1)^4 \in [0,1]$ for $p \in [0,1]$.

All three conditions are satisfied. The classical Lindeberg–Lévy CLT therefore applies.

7.3 Statement of the limit theorem

Let $Y_1, \dots, Y_N$ be i.i.d. with mean $\mu_Y \equiv \mathbb{E}[Y_k] = (2p-1)^2$ and variance $\sigma_Y^2 \equiv \mathrm{Var}(Y_k) = 1 - (2p-1)^4$. The Lindeberg–Lévy CLT states

$$Z_N \;\equiv\; \frac{S - N\mu_Y}{\sigma_Y\sqrt{N}} \;\xrightarrow{d}\; \mathcal{N}(0,1) \qquad \text{as } N \to \infty, \tag{30}$$

where $\xrightarrow{d}$ denotes convergence in distribution. Equivalently, in the original scale,

$$S \;\xrightarrow{d}\; \mathcal{N}\Big(N(2p-1)^2,\;\; N\big[1 - (2p-1)^4\big]\Big). \tag{31}$$

For finite N, Eq. (31) is an *approximation* whose accuracy improves with N. For our parameters ($N = 25$ in `sample_small`, $N = 50$ in `sample_larger`) the Berry–Esseen bound predicts a Kolmogorov distance to the limiting Gaussian of $\mathcal{O}(1/\sqrt{N}) \lesssim 0.2$ in the worst case and substantially better in the bulk – adequate for tail tests at the level we care about, and verified empirically against Task 6 Monte Carlo.

7.4 From the Gaussian density to a probability mass

The CLT in Eq. (31) gives a *continuous* density

$$\varphi(s) \;=\; \frac{1}{\sqrt{2\pi\,\sigma_N^2}}\,\exp\!\left[-\frac{(s - \langle S\rangle)^2}{2\sigma_N^2}\right], \tag{32}$$

but S is discrete. Two structural facts about the lattice on which S lives turn $\varphi(s)$ into the probability mass we actually want:

- **Parity.** Each $Y_k \in \{-1, +1\}$ changes S by ± 1 relative to $-N$ (its minimum), so $S + N$ is always even and therefore S has the same parity as N.
- **Spacing.** Flipping a single Y_k from $+1$ to -1 changes S by exactly 2. Consecutive allowed values of S are separated by $\Delta s = 2$, never by 1.

The probability of landing on a specific allowed value s is the density times the lattice spacing:

$$\mathbb{P}[\,S = s\,] \;\approx\; \varphi(s)\cdot\Delta s \;=\; 2\,\varphi(s). \tag{33}$$

For tail probabilities (which is what we will use in the hypothesis test), the discrete-to-continuous correction is handled by a half-spacing shift,

$$\mathbb{P}[\,S \ge s\,] \;\approx\; 1 - \Phi\!\left(\frac{(s-1) - \langle S\rangle}{\sigma_N}\right), \tag{34}$$

where $\Phi(\cdot)$ is the standard normal CDF and $s-1$ implements the standard continuity correction at the lattice spacing of 2 (half-step $= 1$).

7.5 Closed-form null distribution

Combining Eqs. (32)–(33) with the Task 5 moments gives the final pointwise distribution.

Null distribution of S (homogeneous p)

For two independent students answering N questions, each with common difficulty p,

$$\boxed{\mathbb{P}[\,S = s\,] \;\approx\; \frac{2}{\sqrt{2\pi\,N\big[1-(2p-1)^4\big]}}\;\exp\!\left[-\frac{\big(s-N(2p-1)^2\big)^2}{2\,N\big[1-(2p-1)^4\big]}\right],} \tag{35}$$

on the lattice $s \in \{-N, -N+2, \ldots, N-2, N\}$. The corresponding upper-tail probability is

$$\boxed{\mathbb{P}[\,S \geq s_{\text{obs}}\,] \;\approx\; 1-\Phi\!\left(\frac{s_{\text{obs}}-1-N(2p-1)^2}{\sqrt{N\big[1-(2p-1)^4\big]}}\right).} \tag{36}$$

7.6 Sanity checks at the extremes

The closed form in Eq. (35) reduces correctly in every limit Task 5 already covered, plus one new shape-level check.

regime	$\langle S\rangle$	σ_N^2	shape of $\mathbb{P}[S=s]$
$p=\frac{1}{2}$	0	N	symmetric Gaussian on ± 1 random walk
$p\to 0$ or $p\to 1$	N	0	Dirac mass at $s=N$ (Gaussian collapses)
$N\to\infty$ at fixed p	$\sim N(2p-1)^2$	$\sim N$	relative width $\sigma_N/\langle S\rangle = \mathcal{O}(N^{-1/2})$

Table 6: Limiting behaviour of the null distribution Eq. (35).

The $p \to \frac{1}{2}$ recovery is a textbook check

At $p=\frac{1}{2}$, Eq. (35) simplifies to

$$\mathbb{P}[\,S = s\,]\big|_{p=1/2} \approx \frac{2}{\sqrt{2\pi N}}\exp\!\left[-\frac{s^2}{2N}\right], \tag{37}$$

which is exactly the de Moivre–Laplace approximation to the ± 1 random-walk endpoint distribution after N steps. The random-walk limit was a known result *before* we did any of this; recovering it from a parameter-dependent derivation is the classical sanity check.

7.7 Sharpening the $(5, 10)$ verdict

We can now upgrade Task 5's informal "$\sim 5\sigma_N$" statement into a calibrated p-value. For `sample_small` we use the pooled difficulty estimate $\hat{p} \approx 0.558$ and $N = 25$, so

$$\langle S\rangle \;\approx\; 25\,(2(0.558)-1)^2 \;\approx\; 0.336, \tag{38}$$

$$\sigma_N \;\approx\; \sqrt{25\big[1-(0.116)^4\big]} \;\approx\; 5.000. \tag{39}$$

The observed pair has $s_{\text{obs}} = 25 = N$. Plugging into the continuity-corrected tail formula Eq. (36):

$$\mathbb{P}[\, S \geq 25\,] \;\approx\; 1 - \Phi\!\left(\frac{25 - 1 - 0.336}{5.000}\right) \;=\; 1 - \Phi(4.733) \;\approx\; 1.10 \times 10^{-6}. \tag{40}$$

Correcting for multiple comparisons. We are not testing a *prespecified* pair; we are scanning all $\binom{M}{2} = 1225$ off-diagonal entries of $\boldsymbol{S}$ and picking the largest. A naïve Bonferroni correction inflates the p-value by a factor of 1225:

$$p_{\text{Bonferroni}} \;\approx\; 1225 \cdot 1.10 \times 10^{-6} \;\approx\; 1.35 \times 10^{-3}. \tag{41}$$

Conviction-grade evidence against (5, 10)

Even after the most conservative correction for scanning 1225 candidate pairs, the probability that two *honest* students would achieve $S \geq 25$ on a 25-question test of overall difficulty $\hat{p} \approx 0.558$ is $p_{\text{Bonferroni}} \approx 1.35 \times 10^{-3}$ – roughly one in 740. Compare to a conventional $\alpha = 0.01$ threshold: we reject the null hypothesis of independence at better than the 1% level even *after* accounting for multiple testing.

The qualitative "shortlist" verdict of Task 4 has hardened into a quantitative rejection of the null. The pair $(5, 10)$ is now flagged not just as a geometric outlier in $\boldsymbol{S}$ but as a statistical impossibility under independent test-taking.

7.8 Looking ahead to Task 8

The homogeneous-p derivation above suffices for `sample_small`, where the empirical column means are not too dispersed. For `sample_larger` ($M = 10{,}000$, $N = 50$) we have no reason to expect questions to share difficulty, and the identically-distributed assumption in (ii) of the CLT box fails.

The fix is to drop (ii) and invoke the more general *Lindeberg–Feller CLT*, which only requires (i) independence, (iii) finite variance *plus* a uniform smallness condition on individual variances. Both extensions hold trivially: the $Y_k = X_{ik}X_{jk}$ remain bounded by 1, so the Lindeberg condition is automatic.

The mean and variance generalise term-by-term:

$$\langle S \rangle \;=\; \sum_{k=1}^{N} (2p_k - 1)^2, \qquad \sigma_N^2 \;=\; \sum_{k=1}^{N} \bigl[1 - (2p_k - 1)^4\bigr], \tag{42}$$

where p_k is the empirical column-mean accuracy on question k. Equations (35)–(36) retain their Gaussian form; only the moments are replaced by Eq. (42). Task 8 then computes $z_{ij} = (S_{ij} - \langle S \rangle)/\sigma_N$ for every pair and flags those exceeding a threshold corrected for the $\binom{10{,}000}{2} \approx 5 \times 10^7$ candidate comparisons.

8 Task 8 – Identifying Colluders in `sample_larger`

8.1 Why the null model must become heterogeneous

The homogeneous-p derivation above is adequate for `sample_small`, but `sample_larger` contains $M = 10{,}000$ students and $N = 50$ questions whose empirical difficulties are visibly uneven. In the larger dataset the estimated per-question correct-answer probabilities range from

$$\min_k \hat{p}_k = 0.4903 \qquad \text{to} \qquad \max_k \hat{p}_k = 0.9015,$$

with mean $\bar{p} = 0.531946$. A single pooled p would therefore understate the chance that two independent students agree on the easy questions.

The natural extension is to keep independence across students but allow each question to have its own difficulty $\hat{p}_k$. For a pair of independent students, define

$$Y_k = X_{ik} X_{jk} \in \{-1, +1\}.$$

Then

$$\mathbb{P}(Y_k = +1) = \hat{p}_k^2 + (1 - \hat{p}_k)^2 \equiv q_k, \qquad \mathbb{E}[Y_k] = (2\hat{p}_k - 1)^2. \tag{43}$$

The similarity score is still

$$S_{ij} = \sum_{k=1}^{50} Y_k,$$

but it is now a sum of independent, non-identically distributed Bernoulli-derived variables. Since each Y_k is bounded by 1, the Lindeberg condition is automatic, and the Task 7 Gaussian logic generalises term-by-term:

$$\mu_S = \mathbb{E}[S_{ij}] = \sum_{k=1}^{50} (2\hat{p}_k - 1)^2 = 2.556724, \tag{44}$$

$$\sigma_S^2 = \mathrm{Var}(S_{ij}) = \sum_{k=1}^{50} \left[1 - (2\hat{p}_k - 1)^4\right] = 48.370389, \qquad \sigma_S = 6.954882. \tag{45}$$

8.2 Exact tail calculation

For final decisions we do not need to rely only on the Gaussian approximation. Let K_{ij} be the number of questions on which students i and j agree. Since $S_{ij} = 2K_{ij} - 50$, the event $S_{ij} \geq s$ is equivalent to

$$K_{ij} \geq \frac{s + 50}{2}.$$

Under the heterogeneous null, K_{ij} has the Poisson-binomial distribution with success probabilities $q_1, \ldots, q_{50}$ from Eq. (43). We compute that distribution by dynamic programming:

$$f_r^{(k)} = (1 - q_k) f_r^{(k-1)} + q_k f_{r-1}^{(k-1)}, \qquad f_0^{(0)} = 1.$$

The resulting upper-tail probabilities are shown in Table 7.

threshold s	$\mathbb{P}(S \geq s)$ under null	expected number among $\binom{10{,}000}{2}$ pairs
36	4.7817×10^{-7}	23.906
38	7.9163×10^{-8}	3.958
40	1.0995×10^{-8}	0.550
42	1.2458×10^{-9}	0.062
44	1.1060×10^{-10}	0.0055
46	7.2141×10^{-12}	3.61×10^{-4}
48	3.0743×10^{-13}	1.54×10^{-5}
50	6.4213×10^{-15}	3.21×10^{-7}

Table 7: Heterogeneous-question null tails for `sample_larger`. Even after scanning nearly fifty million pairs, an identical pair ($S = 50$) has expected count far below one.

8.3 Search procedure

The computation is intentionally simple and auditable:

(a) estimate $\hat{p}_k$ from the 10,000 × 50 design matrix;

(b) compute the exact Poisson-binomial null distribution for pairwise agreement counts;

(c) scan all off-diagonal pairwise dot products $S_{ij} = \boldsymbol{X}_{i,\cdot}\boldsymbol{X}_{j,\cdot}^{\top}$;

(d) record every pair with $S_{ij} \geq 40$, a threshold whose null expectation is already below one false positive in the whole scan.

The implementation is in `src/task8.py`; the machine-readable outputs are written to `data/output/task8/tas`
and `data/output/task8/task8_high_similarity_pairs.csv`.

8.4 Result

The scan finds exactly ten pairs with $S \geq 40$, and all ten have the maximum possible similarity $S = 50$. They form one complete five-student clique:

$$\{85,\ 1351,\ 1906,\ 3542,\ 5362\} \quad \text{(zero-based indices)}.$$

Equivalently, if students are numbered from 1 rather than Python's 0,

$$\{86,\ 1352,\ 1907,\ 3543,\ 5363\}.$$

student i	student j	similarity S_{ij}
85	1351	50
85	1906	50
85	3542	50
85	5362	50
1351	1906	50
1351	3542	50
1351	5362	50
1906	3542	50
1906	5362	50
3542	5362	50

Table 8: All high-similarity pairs found in Task 8, using zero-based indexing. There are no other pairs with $S \geq 40$.

Task 8 final answer

The likely colluding group in `sample_larger` is

$$\boxed{\{85,\ 1351,\ 1906,\ 3542,\ 5362\}}$$

in zero-based indexing, or

$$\boxed{\{86,\ 1352,\ 1907,\ 3543,\ 5363\}}$$

in one-based student numbering. Each pair inside this group has $S = 50$, meaning all 50 answers are identical. Under the heterogeneous-question null, a single independent pair has probability 6.4213×10^{-15} of achieving $S = 50$, and the expected number of such pairs among all 49,995,000 comparisons is only 3.21×10^{-7}. This is decisive evidence of coordinated answering rather than chance agreement.

8.5 Interpretation

The conclusion is stronger than the small-sample $(5, 10)$ result. In Task 2 the suspicious pair was an isolated maximum on a 25-question test. Here, five students share the same complete 50-question answer vector, producing a ten-edge clique at the absolute maximum possible similarity. The heterogeneous null model explicitly accounts for easy questions, so the result cannot be explained away by the test having high-accuracy columns. The only statistically coherent interpretation is that these five answer sheets were not generated independently.